

# Specific Aims

## Project Title:

*Targeting Epigenetic Drift in Cellular Aging: A Pilot Human Study of Ergothioneine (EGT)*

**Aging is the single greatest risk factor for nearly all chronic diseases.** Among the hallmarks of aging, *epigenetic drift*—the gradual loss of fidelity in DNA methylation patterns—has emerged as both a marker and potential driver of cellular dysfunction and age-related disease. While DNA methylation clocks now enable quantitative tracking of biological age, few interventions have been shown to slow or reverse these changes in healthy humans.

**Ergothioneine (EGT)** is a naturally occurring amino acid derivative with potent antioxidant and cytoprotective properties. It is acquired exclusively through diet and transported by the highly selective **OCTN1 transporter (SLC22A4)**, which is enriched in mitochondria and long-lived tissues. Emerging evidence suggests that EGT plays a protective role against age-related cellular damage, including oxidative stress, mitochondrial dysfunction, and telomere shortening. However, its role in regulating **epigenetic aging in humans has never been explored.**

Given its excellent safety profile and status as a dietary supplement, **EGT is uniquely positioned for rapid clinical translation**—a compelling opportunity for aging therapeutics.

## Central Hypothesis

We hypothesize that **oral supplementation with EGT reduces the rate of epigenetic drift in healthy middle-aged adults**, measurable via DNA methylation clocks over a 3-month period.

## Aim 1. Evaluate the impact of EGT on biological aging using DNA methylation clocks in humans.

We will conduct a randomized, placebo-controlled pilot trial (N=20–30, 3 months) in healthy middle-aged adults. Peripheral blood will be collected at baseline and post-intervention for targeted methylation profiling (Horvath, PhenoAge, GrimAge). The **primary mechanistic endpoint** will be change in epigenetic age acceleration relative to placebo. Plasma EGT levels and OCTN1 genotype will be assessed as pharmacodynamic anchors.

## Aim 2. Assess EGT's impact on complementary geroscience biomarkers

We will measure circulating markers of oxidative stress (8-OHdG, GSH/GSSG), inflammation (IL-6, CRP, TNF- $\alpha$ ), and SIRT6 expression/activity. These serve as a “**speedometer panel**” of mechanistic engagement and provide multidimensional evidence beyond clocks alone.

## Aim 3. Establish feasibility, safety, and exploratory functional outcomes

We will evaluate adherence (pill counts, plasma EGT), safety (laboratory panels, adverse events), and exploratory functional endpoints including grip strength and the SF-36 quality-of-life survey. This ensures translational relevance and feasibility metrics for powering future NIH-scale trials. *If Michael Snyder could be included, he may have a sophisticated mechanism to study this, which would be a very nice set of datapoints to gather, although, a simple smart dispenser would be enough and cheap.*

## Innovation & Significance

- **Novel Mechanism:** This is the first study to test *epigenetic aging modulation* by EGT in humans.
- **High Translational Potential:** EGT is available as a dietary supplement and can rapidly enter broader clinical trials.
- **Feasibility:** The study leverages sensitive methylation clocks, requires minimal infrastructure, and is executable within seed-level funding.

## Impact

This pilot study directly addresses the high-risk, high-reward mandate of the Rosenkranz Seed Grant by testing a *safe, inexpensive, and understudied small molecule* in the most clinically relevant model—humans. Positive findings would lay the groundwork for a new class of longevity interventions and justify expanded NIH funding.

## Does this project challenge a prevailing assumption?

This project challenges the prevailing assumption that safe, naturally occurring dietary compounds cannot modulate core hallmarks of aging in healthy humans within clinically practical timeframes. While vitamin D and NMN have offered intriguing signals, their effects remain inconsistent. By focusing on ergothioneine—a compound with unique mitochondrial transport and mechanistic ties to SIRT6—we directly test whether epigenetic drift, a hallmark long thought to be intractable to dietary intervention, can be slowed measurably in just three months. A positive signal would reframe the potential of nutraceuticals in geroscience and catalyze a new class of translational trials.

## Endpoints

From [this](#) paper

- Biomarkers associated with both alzhiemers and aging are good ones if they could be argued to be novel. For alzhiemers see [here](#).
- Response biomarkers are also good because they show the body actually did have some response to a therapeutic, although this would be a secondary thing I suppose... It seems hard to quantify age specific ones
- Epigenetic clocks are noted as increasingly recognized surrogate response biomarkers. Perhaps a good end point? This lines up with Horvath's work (he first proposed these).
- There is a table (2) of clocks. Maybe the Aging.AI one is good but it seems it is only licensed for clinical trials (it does seem to have a comprehensive set of markers, review them).
- See some studies that use supplements for examples. For example the **"supplement" section of table 3**, which seems to be perfect
- Standard of academically recognized geroprotector is high. Only a few are proposed and include things like calorific diet moderation, regular exercise, things like this.

## Graveyard

Question would be, is it possible to study methylation clocks in the brain by imaging? If so that is one means to study anti-aging effect on the brain due to EGT. It is said to be challenging, perhaps Michelle James has some thoughts on this if she were PI, but there are other means to measure the impact of EGT on the brain (alzhiemers)

EGT is also broadly uptaken in many tissue, so methylation based biomarkers should be readily studied over a 3-12 month period

Make sure to select a biomarker that is a “speedometer” as this is the current rate of aging and not the biological age. That would be a good normalization because any delta in these speedos is a positive outcome see [here](#) and [here](#) (first one is probably a bit more useful)

Note that different cells age differently and different cells may be sensitive to EGT more or less so. You have to be careful where you sample to have the best chance of getting positive results.

PI's

I looked at three: Ane Brunet (chair, cannot sponsor) Michael Snyder (wearables focus, not sure how to appeal) Katrin Faye Chua (laser focus on SIRT6, does not really serve as an end point and unclear how this fits in with her research).